

UniForge

Blender to Unity Asset Bridge

Technical Specification & Product Design Document

Format: .unif — Universal Node-based Interchange Format

Version 1.0 Specification

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1. Project Overview

UniForge is a two-part tool that enables seamless asset transfer between Blender and Unity, preserving Shader Node Graphs as Unity Shader Graph materials. It introduces the .unif (Universal Node-based Interchange Format) file format as a clean, focused interchange container for 3D assets, materials, and shader data.

The problem UniForge solves: existing export pipelines (FBX, GLTF) either discard Blender Shader Node Graphs entirely or require extensive manual recreation of materials in Unity. UniForge bridges this gap by reading, encoding, and faithfully reconstructing shader logic across both platforms.

Format	.unif — Universal Node-based Interchange Format
Addon Name	UniForge
Components	Blender Python Addon (Exporter) + Unity C# Plugin (Importer)
Target Users	Game developers, 3D artists, indie developers
Distribution	Blender Market, Unity Asset Store, Gumroad
License	Commercial — Paid, Single Developer License

2. Goals & Motivation

2.1 Primary Goals

- Export Blender Shader Node Graphs and reconstruct them as Unity Shader Graph materials
- Preserve mesh geometry, UVs, normals, transforms, and material slot assignments
- Provide a single .unif file that contains everything Unity needs, nothing it does not
- Support the most common PBR node types covering 90%+ of real-world use cases
- Be usable by solo indie developers with minimal technical knowledge

2.2 Non-Goals (v1.0)

- Displacement / Cycles-only nodes (Light Path, Volume, SSS)
- Skeletal animations and armatures (planned v1.5)
- Bi-directional sync (Unity back to Blender)
- Real-time live link between applications
- Support for Unreal Engine, Godot, or other engines (planned v2.0+)

2.3 Market Gap

No existing tool on Blender Market or Unity Asset Store provides true Shader Node Graph transfer. Developers currently spend hours manually recreating materials after export. UniForge eliminates this workflow bottleneck entirely.

3. The .unif File Format

The .unif format is a structured text-based format (JSON-like, custom syntax) designed to be human-readable during development and parseable by both Python and C#. A binary version (.unifb) is planned for v2.0 to reduce file size.

3.1 Format Structure

Every .unif file consists of named blocks enclosed in square brackets. Each block contains key-value pairs or nested sub-blocks.

Example .unif file structure:

```
[UNIF]
  version: 1.0
  generator: UniForge Blender Addon 1.0
  source_file: tiles.blend

[MESH]
  name: FloorTile
  vertices: [0.5,0.0,0.5, -0.5,0.0,0.5, ...]
  faces: [0,1,2, 2,3,0, ...]
  uvs: [0.0,0.0, 1.0,0.0, ...]
  normals: [0.0,1.0,0.0, ...]

[TRANSFORM]
  position: 0.0, 0.0, 0.0
  rotation: 0.0, 0.0, 0.0
  scale: 1.0, 1.0, 1.0

[MATERIAL]
  name: WetTile_Mat
  slot: 0
  [NODE PrincipledBSDF id=0]
    roughness: 0.35
  [NODE ImageTexture id=1 path=tiles_diffuse.png]
  [CONNECTION]
    1.Color -> 0.Base_Color
```

3.2 Embedded Textures

By default, textures are referenced by relative path. Optionally, textures can be Base64-encoded and embedded directly in the .unif file using the [TEXTURE_EMBEDDED] block. This creates a self-contained single-file workflow at the cost of larger file size.

4. Blender Addon (Exporter)

The Blender component is a Python addon registered as a standard Blender plugin. It appears in the File > Export menu as 'UniForge Asset (.unif)' and in the N-Panel sidebar under a dedicated UniForge tab.

4.1 Installation

- Standard Blender addon installation via Edit > Preferences > Add-ons > Install
- Compatible with Blender 3.6 LTS and Blender 4.x
- No external Python dependencies required

4.2 Export Dialog Options

When the user triggers export, a file dialog opens with the following options:

Option	Default	Description
Export Selection Only	False	Export only selected objects or entire scene
Embed Textures	False	Base64-encode textures into the .unif file
Bake Unsupported Nodes	True	Auto-bake nodes without Unity equivalent to texture
Apply Modifiers	True	Apply all Blender modifiers before exporting mesh
Coordinate System	Unity (Y-up)	Auto-converts Blender Z-up to Unity Y-up axis

4.3 Exporter Internal Pipeline

The exporter follows this pipeline for each object:

1. Validate object: must be MESH type with at least one material slot
2. Apply modifiers (if enabled) via evaluated depsgraph
3. Triangulate mesh and extract vertices, faces, UVs, normals
4. Convert coordinate system (Z-up to Y-up, flip triangle winding)
5. Extract transform (location, rotation_euler, scale)
6. For each material slot: walk node tree, identify node types
7. Supported nodes: serialize to .unif NODE blocks
8. Unsupported nodes: auto-bake to PNG texture if Bake option enabled
9. Serialize all connections between nodes
10. Write complete .unif file to disk

5. Unity Plugin (Importer)

The Unity component is a C# Editor plugin distributed as a Unity Package (.unitypackage). It registers a custom AssetImporter for .unif files, meaning any .unif file dropped into the Assets folder is automatically processed by UniForge.

5.1 Installation

- Import via Window > Package Manager > Import Custom Package
- Compatible with Unity 2021.3 LTS, Unity 2022.x, Unity 6
- Supports both URP and HDRP render pipelines
- Requires Shader Graph package (included by default in URP/HDRP projects)

5.2 Auto-Import Behavior

When a .unif file is detected in the Assets folder, UniForge automatically:

- Parses the .unif file header and validates version compatibility
- Creates a Unity Mesh asset from geometry data
- Creates a Unity Shader Graph asset reconstructing the node network
- Creates a Unity Material asset referencing the generated shader
- Creates a Unity Prefab combining mesh and material
- Places all generated assets in a sub-folder named after the .unif file

5.3 Importer Internal Pipeline

11. Parse .unif file: read all blocks into internal data structures
12. Build Mesh: populate UnityEngine.Mesh with vertices, triangles, UVs, normals
13. Build Shader Graph: instantiate nodes via ShaderGraph API, set parameters
14. Connect nodes: recreate all connections from CONNECTION blocks
15. Create Material: assign generated shader
16. Create Prefab: MeshRenderer + MeshFilter + material
17. Report: show import summary with any warnings about unsupported nodes

5.4 UniForge Inspector Panel

Imported .unif assets show a custom Inspector panel with: source file info, node count, texture list, re-import button, and a visual node graph preview.

6. Supported Node Types

The following tables list all Blender shader nodes and their conversion status in UniForge v1.0.

6.1 Output & Shader Nodes

Blender Node	Unity Equivalent	Status
Material Output	Master Stack (Fragment)	Supported
Principled BSDF	Lit (PBR) Node	Supported
Emission	Emission Node	Supported
Mix Shader	Lerp / Alpha Blend	Partial
Add Shader	Add Blend	Partial
Transparent BSDF	Alpha Clip / Fade	Supported
Glass BSDF	Custom HLSL	Partial
Subsurface Scattering	N/A	Bake Only
Volume Absorption	N/A	Not Supported
Volume Scatter	N/A	Not Supported

6.2 Texture Nodes

Blender Node	Unity Equivalent	Status
Image Texture	Sample Texture 2D	Supported
Normal Map	Normal Unpack Node	Supported
Noise Texture	Simple Noise Node	Supported
Voronoi Texture	Voronoi Noise Node	Supported
Musgrave Texture	Gradient Noise	Partial
Wave Texture	Custom HLSL	Partial
Gradient Texture	Gradient Node	Supported
Checker Texture	Checkerboard Node	Supported
Brick Texture	N/A	Bake Only
Environment Texture	Cubemap Sample	Partial

6.3 Color & Math Nodes

Blender Node	Unity Equivalent	Status
Mix RGB	Blend Node	Supported
Math	Math Node (all ops)	Supported
Vector Math	Vector Math Node	Supported
Color Ramp	Gradient Node	Supported
Hue/Saturation/Value	HSV Node	Supported
Bright/Contrast	Contrast Node	Supported
Gamma	Power Node	Supported
Invert	One Minus Node	Supported
RGB Curves	Custom HLSL Curve	Partial

6.4 Input / Coordinate Nodes

Blender Node	Unity Equivalent	Status
UV Map	UV Node	Supported
Texture Coordinate	UV / Position / Normal	Supported
Mapping	Transform UV Node	Supported
Vertex Color	Vertex Color Node	Supported
RGB	Color Node	Supported
Value	Float Node	Supported
Fresnel	Fresnel Node	Supported
Layer Weight	Fresnel Node	Partial
Light Path	N/A	Not Supported
Object Info	Object Position (partial)	Partial

7. Versioning & Roadmap

Version	Target Date	Features
v1.0	Q3 2026	Core export: Mesh, UV, Normals, Transforms, PBR Nodes, Image Textures, Node Connections, Material Slots, URP + HDRP support
v1.1	Q4 2026	Embedded texture support, Batch export (multiple objects), UniForge Inspector panel, Import report/log
v1.2	Q1 2027	Partial node bake fallback improvements, Custom HLSL generation for Wave/RGB Curves, Progress bar for large scenes
v1.5	Q2 2027	Skeletal armature export, Keyframe animation export, Shape Keys / Morph Targets, Action clip export
v2.0	Q4 2027	Binary .unifb format, LOD export, Collision mesh export, Godot Engine support (experimental), CLI export tool
v2.5	2028+	Godot stable support, Unreal Engine 5 experimental, Real-time live-link mode (Blender <-> Unity viewport sync)

8. Distribution & Monetization

8.1 Pricing Strategy

Product	Price	Includes
UniForge Complete	€29.99	Blender Addon + Unity Plugin + lifetime updates for major version
UniForge Blender Only	€14.99	Blender Addon only (.unif exporter)
UniForge Studio License	€79.99	Up to 5 developers, priority support, early access to beta versions

8.2 Distribution Channels

- Blender Market — primary channel for Blender Addon, large existing audience
- Unity Asset Store — for Unity Plugin component, high discoverability
- Gumroad — direct sales, higher margin, good for bundles and updates
- GitHub — public issue tracker, documentation, optional open-source lite version

8.3 Marketing

- TikTok / YouTube Shorts: before/after workflow comparison (10 min manual vs 10 sec UniForge)
- Blender Artists forum post with free demo file
- Reddit: r/blender, r/Unity3D, r/gamedev
- Target keywords: blender to unity shader, blender unity material export, shader graph transfer

9. Technical Requirements

9.1 Blender Addon Requirements

Blender Version	3.6 LTS or higher (4.x recommended)
Python Version	3.10+ (bundled with Blender, no separate install)
External Dependencies	None
Blender API Used	bpy.types.Operator, bpy.props, ShaderNodeTree traversal
Platforms	Windows, macOS, Linux (all platforms Blender supports)

9.2 Unity Plugin Requirements

Unity Version	2021.3 LTS or higher
Render Pipeline	URP (primary), HDRP (secondary), Built-in (limited)
Required Packages	Shader Graph (com.unity.shadergraph)
C# Version	C# 9.0+
Platforms	Windows Editor, macOS Editor, Linux Editor

9.3 Development Stack

- Blender Addon: Python 3.10, bpy API, standard library only
- Unity Plugin: C# 9, UnityEditor namespace, ShaderGraph API, ScriptedImporter
- Version Control: Git, public GitHub repository
- Testing: Blender 3.6 + 4.x, Unity 2021 LTS + Unity 6, URP + HDRP
- Documentation: Markdown on GitHub + PDF spec (this document)